

What is Swarm? tl;dr

UNSTOPPABLE STORAGE

Swarm is a **decentralised storage and communication network** run by peer-to-peer **Bee nodes**. Upload data, get back a **content hash** — a permanent address derived from the bytes themselves. Anyone with the hash can retrieve the content from any Bee node or public gateway. Storage is bought in the form of **postage stamps**: prepaid storage batches purchased with **xBZZ** on Gnosis Chain. Batches have a TTL, and will eventually expire unless topped up with more xBZZ. If all batches associated with uploaded content expire, the content will be pruned.

Homepage



Swarm
ethswarm.org

Docs



Documentation
docs.ethswarm.org

Explorer



SwarmScan
swarmscan.io

What you can build on Swarm

BUILD WITH FREEDOM · SCALE WITHOUT LIMITS

File & media storage

Fixed, content-addressed URLs for any file — images, video, PDFs, datasets, ML artifacts, NFT assets.

Live video & streaming

Early proof-of-concept work on HLS over Swarm exists — expect to integrate and tune it yourself.

Decentralised websites

Publish a static front-end straight to the network. No server, no CDN, no DNS lock-in.

Chat, comments & messaging

Real-time chat, threaded comments and peer messaging. Real-time delivery needs a service node with a GSOC subscription (See "Messaging and Chat" on back page).

Blogs & dashboards

Stable URLs whose contents can update. Personal blogs, public dashboards, status pages.

dApp front-ends

Host a static front-end for an Ethereum dApp on Swarm. Pair on-chain contracts with content-addressed UI for a fully decentralised stack.

What Swarm is *not*

- **Not a database.** Opaque blobs addressed by hash. No queries, no indexing, no search — model structure in the app layer.
- **Not permanent without payment.** Content lives only as long as a valid postage stamp covers its chunks.
- **Not instantly consistent.** Uploads propagate asynchronously. Default returns before data is fully pushed (deferred: `true`) — fast for the uploader, content becomes available shortly after. Use `deferred: false` when the next request must read the bytes back immediately (live previews, sync-then-share).
- **Not free to write.** Every upload needs a stamp funded with xBZZ. Reads have a limited free tier on the public gateway.
- **Not a smart-contract platform.** Swarm stores data; it does not execute code. Complements Ethereum, does not replace it.

Limitations & gotchas

- **Stamp batch expiry.** Once a batch expires, its chunks become eligible for pruning — not deleted instantly, but they can disappear at any time after expiry. Stamps expire based on a **dynamic pricing mechanism**, so it's vital to monitor batches and top up if the price rises to ensure data stays online..
- **Ultra-light nodes can't upload.** Writing requires a light node with funds.
- **Feed writes are append-only.** Each index slot writes once; gaps hide newer updates. Always resolve the current index before writing.
- **Feed slots hold ≤ 4 KB.** For larger payloads, upload as a file first then write the resulting reference into the feed.
- **Capacity & lifetime are non-deterministic.** Both depend on chunking and dynamic pricing — monitor and top up or .

QUICKSTART FOR SWARM DEVELOPMENT

THE RECOMMENDED WAY TO START BUILDING · ~5 MIN

1 Install Claude Code

MACOS / LINUX / WSL

```
$ curl -fsSL https://claude.ai/install.sh | bash
```

Or `$ brew install --cask claude-code`

WINDOWS · POWERSHELL

```
$ irm https://claude.ai/install.ps1 | iex
```

Or `$ winget install Anthropic.ClaudeCode`

Source: code.claude.com/docs/en/quickstart

2 Add swarm-quickstart-skills

LOG IN TO CLAUDE, THEN CLONE & OPEN

```
$ claude auth login
→ sign in via browser with OAuth
```

```
$ git clone https://github.com/ethersphere\
/swarm-quickstart-skills.git
```

```
$ cd swarm-quickstart-skills/.claude
$ claude
```

Pre-req: be signed in to claude.ai in your browser, then run `claude auth login` before running `claude`.

3 Run `/swarm` to set up your node

Run `/swarm` to kick off automated setup. It checks for existing installations and installs and configures a new Bee node if needed.

↳ Suggested Developer Journey

```
/swarm → /setup-bee-interactive → /stamps →
/upload-download | /host-website | /blog
```

Try `/upload-download`, `/host-website` & `/blog` for common use cases. Use `/menu` for the full list.

Prefer written docs? → justdeploy.bzz.link

Step-by-step guides for getting started on Swarm — Bee node install, funding, site hosting and more, with tracks for beginner, intermediate and advanced users. Mirror: justdeploy.eth.limo

▸ More on the back: developer URLs, examples, network tools, community ▸

Where to go next scan by category · all links stay updated online



SWARM DEVELOPER RESOURCES

docs.ethswarm.org/docs/develop/resources

One curated landing page with links to every category below. Bookmark it — links stay updated online so this printed sheet keeps working.



LEARN

Get Started

your first 30 minutes



- What is Swarm · </docs/concepts/what-is-swarm>
- Developer introduction · </docs/develop/introduction>
- Bee node quickstart · </docs/bee/installation/quick-start>
- Full installation guide · </docs/bee/installation/getting-started>
- Swarm papers · papers.ethswarm.org

SCAN QR → FULL GET STARTED HUB

Key Concepts

how it actually works



- DISC · </docs/concepts/DISC>
- Incentives & BZZ · </docs/concepts/incentives/overview>
- Postage stamps · </docs/develop/tools-and-features/buy-a-stamp-batch>
- Feeds · </docs/develop/tools-and-features/feeds>
- Manifests · </docs/develop/tools-and-features/manifests>
- Chunk types · </docs/develop/tools-and-features/chunk-types>

SCAN QR → DEEP-DIVE ON CONCEPTS

Developer Guides

recipes for common builds



- Upload & download · </docs/develop/upload-and-download>
- Host a website · </docs/develop/host-your-website>
- Files & directories · </docs/develop/files>
- Routing for SPAs · </docs/develop/routing>
- Dynamic content (feeds) · </docs/develop/dynamic-content>
- Multi-author blog · </docs/develop/multi-author-blog>
- Access control (ACT) · </docs/develop/act>

SCAN QR → ALL DEVELOPER GUIDES

EXAMPLES

Publishing & Websites

static sites, blogs, file uploads



- upload-react · github.com/ethersphere/examples-js/tree/master/upload-react
- Simple Blog · github.com/ethersphere/examples/tree/master/simple-blog
- Basic Static Website · github.com/ethersphere/examples/tree/main/basic-static-website

SCAN QR → MORE PUBLISHING EXAMPLES

Streaming

early HLS-over-Swarm work



- Swarm Stream React Example · github.com/Solar-Punk-Ltd/swarm-stream-react-example
- Swarm Stream JS Library · github.com/Solar-Punk-Ltd/swarm-stream-js-lib
- Reference: feeds · </docs/develop/dynamic-content>

SCAN QR → STREAMING REPO ON GITHUB

Messaging & Chat

PSS & GSOC patterns



- Swarm Chat React Example · github.com/Solar-Punk-Ltd/swarm-chat-react-example
- Swarm Chat JS Library · github.com/Solar-Punk-Ltd/swarm-chat-js
- PSS messaging · </docs/develop/tools-and-features/pss>
- GSOC messaging · </docs/develop/tools-and-features/gsoc>

SCAN QR → CHAT REPO ON GITHUB

TOOLS

SDKs & APIs

talk to Bee from your code



- bee-js · bee-js.ethswarm.org
- Bee HTTP API · docs.ethswarm.org/api
- Swarm CLI · </docs/bee/working-with-bee/swarm-cli>
- Swarm MCP · github.com/ethersphere/swarm-mcp
- Bee Go (source) · github.com/ethersphere/bee

SCAN QR → BEE-JS DOCS

Gateways, Deploy & Network

ship & inspect



- Beeport · app.ethswarm.org
- Public Gateway · gateway.ethswarm.org
- Swarm Gateway · <https://github.com/ethersphere/swarm-gateway>
- Local test network · </docs/develop/tools-and-features/starting-a-test-network>
- SwarmScan · swarmscan.io

SCAN QR → DEPLOY & GATEWAY DOCS

Community & Support

stuck? ask here first



- Discord · discord.ethswarm.org
- Docs home · docs.ethswarm.org
- Bee FAQ · </docs/bee/bee-faq>
- GitHub org · github.com/ethersphere
- Swarm Foundation blog · blog.ethswarm.org

SCAN QR → JOIN DISCORD